WORKED SOLUTION

STEP I 2002, Question 11

Mechanics • Oblique Elastic Collisions

STEP I 2002, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

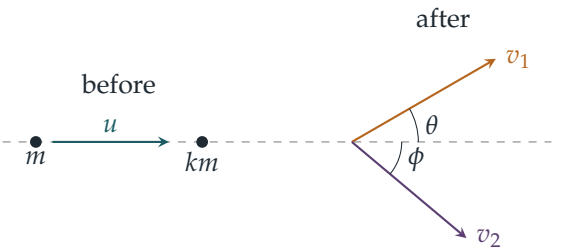
step.maths.org/questions/02-s1-q11

The key idea

A collision gives us two conservation laws: momentum (a vector, so two component equations) and, because no energy is lost, kinetic energy. The two momentum equations are linear in the unknown speeds after impact, so we can solve them for those speeds in terms of the angles. Substituting into the energy equation then removes every speed and leaves a relation between the angles and k alone.

Solution

We model P_1 and P_2 as particles moving in a plane, with no external forces acting during the collision. Let P_1 approach with speed $u > 0$ along the x-axis, and let the speeds after the collision be v_1 (for P_1, at angle θ above the axis) and v_2 (for P_2, at angle ϕ below it), with $0 < \theta < \pi$ and $0 < \phi < \pi$.



The identity

Conservation of momentum along the original line of motion, and perpendicular to it, gives (after cancelling m)

$$v_1 \cos\theta + kv_2 \cos\phi = u,$$
$$v_1 \sin\theta - kv_2 \sin\phi = 0,$$

where the minus sign is there because P_2 moves off on the other side of the axis. Conservation of kinetic energy gives, after cancelling $\frac{1}{2}m$,

$$v_1^2 + kv_2^2 = u^2. \qquad (*)$$

The momentum equations are two linear equations in the two unknowns v_1 and kv_2. Multiply the first by $\sin\phi$ and the second by $\cos\phi$ and add: the kv_2 terms cancel and

$$v_1(\sin\theta\cos\phi + \cos\theta\sin\phi) = u\sin\phi, \qquad \text{so} \qquad v_1 = \frac{u\sin\phi}{\sin(\theta+\phi)}.$$

Similarly, multiplying the first by $\sin\theta$ and the second by $\cos\theta$ and subtracting gives

$$kv_2 = \frac{u\sin\theta}{\sin(\theta+\phi)}.$$

Ask yourself Are we allowed to divide by $\sin(\theta+\phi)$?

We need $\theta + \phi \neq \pi$ (it lies strictly between 0 and 2π). If $\theta + \phi = \pi$, then $\sin\phi = \sin\theta$ and $\cos\phi = -\cos\theta$. The second momentum equation becomes $(v_1 - kv_2)\sin\theta = 0$, so $v_1 = kv_2$ because $\sin\theta \neq 0$. The first then reads $(v_1 - kv_2)\cos\theta = u$, giving $0 = u$, which is impossible. So $\sin(\theta+\phi) \neq 0$ and the division is fine.

Now we substitute into $(*)$, writing kv_2^2 as $(kv_2)^2/k$:

$$\frac{u^2\sin^2\phi}{\sin^2(\theta+\phi)} + \frac{u^2\sin^2\theta}{k\sin^2(\theta+\phi)} = u^2.$$

Multiplying through by $k\sin^2(\theta+\phi)/u^2$ (which is non-zero) gives

$$\sin^2\theta + k\sin^2\phi = k\sin^2(\theta+\phi),$$

as required.

(Geometrically, the momentum equations say that the momenta mv_1, kmv_2 and mu form a triangle with angles ϕ, θ and $\pi - (\theta+\phi)$, and the two expressions above are just the sine rule in that triangle.)

Right angle between the paths

If the paths are at right angles, then $\theta + \phi = \frac{\pi}{2}$, so $\sin^2(\theta+\phi) = 1$ and $\sin\theta = \cos\phi$. The identity becomes

$$\cos^2\phi + k\sin^2\phi = k, \qquad \text{that is,} \qquad \cos^2\phi = k(1-\sin^2\phi) = k\cos^2\phi.$$

Here $\cos\phi \neq 0$: otherwise $\phi = \frac{\pi}{2}$ and then $\theta = 0$, but the question tells us θ is non-zero. Dividing by $\cos^2\phi$ gives $k = 1$, as required.

Checks

- The familiar fact from snooker: when equal masses collide elastically with one at rest, they move off at right angles. Our result is the converse.
- A numerical example: take $k = 2$ and $\phi = 30°$. A smooth elastic impact along that line sends P_2 off at speed $\frac{2}{3}u\cos 30°$, and P_1 then moves at $\theta = 90°$. Both sides of the identity equal $\frac{3}{2}$.
- We also simulated 2×10^5 random elastic collisions (random k, u and line of impact). The identity and both formulas for v_1 and kv_2 held to within 10^{-12}.

Common slips

- Writing the perpendicular momentum equation as $v_1 \sin\theta + kv_2 \sin\phi = 0$: the particles move off on opposite sides, so the components have opposite signs.
- Forgetting the k in the energy equation, or failing to say why $\cos\phi \neq 0$ in the final step.

Beyond STEP

In the frame moving with the centre of mass, an elastic collision just rotates both velocities, keeping their sizes. All of these angle relations come from mapping that circle of possible outcomes back to the laboratory frame. This viewpoint is the standard way to treat scattering, from billiard balls to particle physics. In particle physics, measured scattering angles are used to work out the mass ratio of the colliding particles. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops the centre-of-mass treatment of collisions carefully.

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